

Transistoria ISA

General Specifications

Architecture: RISC

Instruction Size: 8 Bit

Maximum Program Size: 16 Instructions (rollback to 0, if exceeded)

Word Width: 4 Bit

Registers: in (000), A (001), B (010), C (011), D (100), E (101), F (110), out (111)

(‘in’ and ‘out’ are I/O ports and writes to ‘in’ do nothing)

Instruction Overview

Mnemonic	7	6	5	4	3	2	1	0	ALU Operation?	Description
NOP	0	0	0	0	0	0	0	0	No	Does nothing
HLT	0	0	0	1	0	0	0	0	No	Halts program
MOV	0	0	1	0	Y	R	R	R	No	If Y is 0, writes content of A into R, writes content of R into A otherwise
ADD	0	0	1	1	0	R	R	R	Yes	Adds values of A and B and puts the result into R
SUB	0	1	0	0	0	R	R	R	Yes	Subtracts value of B from value of A and puts the result into R
AND	0	1	0	1	0	R	R	R	Yes	Bitwise AND with values of A and B, writes result into R
NOR	0	1	1	0	0	R	R	R	Yes	Bitwise NOR with values of A and B, writes result into R
XOR	0	1	1	1	0	R	R	R	Yes	Bitwise XOR with values of A and B, writes result into R
RSH	1	0	0	0	0	R	R	R	Yes	Writes right shifted value of A into R
LDI	1	0	0	1	A	A	A	A	No	Loads 4-bit immediate A into register A
ADI	1	0	1	0	A	A	A	A	Yes	Adds 4-bit immediate to register A
JMP	1	0	1	1	A	A	A	A	No	Jumps to address A
JIZ	1	1	0	0	A	A	A	A	No	Jumps to address A, if last ALU operation resulted in 0
JIC	1	1	0	1	A	A	A	A	No	Jumps to address A, if last ALU operation carried out 1
JNZ	1	1	1	0	A	A	A	A	No	Jumps to address A, if last ALU operation didn't result in 0
JNC	1	1	1	1	A	A	A	A	No	Jumps to address A, if last ALU operation carried out 0

VM Format

Each instruction must consist of two hexadecimal digits, that represent its binary representation. After every instruction, that is followed by another instruction, there must be exactly one new line, before the next instruction.

Examples

91

22

9F

41

E3

10

Load 1 into A

Move A to B

Load 15 into A

Subtract 1 from A

Jump to 3rd (zero-based indexing) instruction, if it didn't result in a 0.

Stop the program otherwise